

The Coin Change Problem

Problem

Submissions

Leaderboard

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Given an amount and the denominations of coins available, determine how many ways change can be made for amount. There is a limitless supply of each coin type.

Example

$n = 3$

$c = [8, 3, 1, 2]$

There are **3** ways to make change for $n = 3$: $\{1, 1, 1\}$, $\{1, 2\}$, and $\{3\}$.

Function Description

Complete the getWays function in the editor below.

getWays has the following parameter(s):

- int n : the amount to make change for
- int $c[m]$: the available coin denominations

Returns

- int: the number of ways to make change

Input Format

The first line contains two space-separated integers n and m , where:

n is the amount to change

m is the number of coin types

The second line contains m space-separated integers that describe the values of each coin type.

Constraints

- $1 \leq c[i] \leq 50$
- $1 \leq n \leq 250$
- $1 \leq m \leq 50$
- Each $c[i]$ is guaranteed to be distinct.

Hints

Solve overlapping subproblems using [Dynamic Programming \(DP\)](#):

You can solve this problem recursively but will not pass all the test cases without optimizing to eliminate the [overlapping subproblems](#). Think of a way to store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for **0** cents? - How many ways can you make change for **> 0** cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case ($n = 0$). - The answer may be larger than a **32**-bit integer.

Sample Input 0

```
4 3
1 2 3
```

Sample Output 0

Explanation 0

There are four ways to make change for $n = 4$ using coins with values given by $C = [1, 2, 3]$:

1. $\{1, 1, 1, 1\}$
2. $\{1, 1, 2\}$
3. $\{2, 2\}$
4. $\{1, 3\}$

Sample Input 1

```
10 4
2 5 3 6
```

Sample Output 1

```
5
```

Explanation 1

There are five ways to make change for $n = 10$ units using coins with values given by $C = [2, 5, 3, 6]$:

1. $\{2, 2, 2, 2, 2\}$
2. $\{2, 2, 3, 3\}$
3. $\{2, 2, 6\}$
4. $\{2, 3, 5\}$
5. $\{5, 5\}$

[Change Theme](#)

Language

Python 3



```

1  #!/bin/python3
2
3  import math
4  import os
5  import random
6  import re
7  import sys
8  #
9  # Complete the 'getWays' function below.
10 #
11 # The function is expected to return a LONG_INTEGER.
12 # The function accepts following parameters:
13 # 1. INTEGER n
14 # 2. LONG_INTEGER_ARRAY c
15 #
16 def getWays(n, c):
17     # Write your code here
18     dp = [[0]*(n+1)] * (len(c)+1)
19     dp[0][0]=1
20     for i in range(1,len(c)+1):
21         for j in range(n+1):
22             if c[i-1]>j:
23                 dp[i][j]=dp[i-1][j]
24             else:
25                 dp[i][j]=dp[i-1][j]+dp[i][j-c[i-1]]
26     return dp[len(c)][n]
```

```
26         return op[-1][-1]
27     if __name__ == '__main__':
28         fptr = open(os.environ['OUTPUT_PATH'], 'w')
```

Line: 37 Col: 1

 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

✔ Sample Test case 1

Input (stdin)

Download

1	4 3
2	1 2 3

Your Output (stdout)

1	4
---	---

Expected Output

Download

1	4
---	---